UPEN

Career: Engineering in Information Technologies and Digital Innovation

Subject: Object-Oriented Programming

Activity: Justification of the class diagram

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Justification of the class diagram

Vehicle Rental Management System

This document details the technical justification for the object-oriented class diagram design developed for the vehicle rental system. The conceptual model has been structured according to the principles of modularity, encapsulation, and data normalization to ensure the system's scalability, consistency, and transactional integrity.

1. Analysis and Justification of Classes and Attributes

Client Class

Purpose: Represents the users or entities that carry out rental transactions within the system.

Attributes and justification:

- **DNI: string:** Functions as an official identifier or legal identity document, ensuring the legal uniqueness of the client.
- **name: string:** Stores the full name of the client for identification and contact purposes.
- **address: string:** Physical location data required for billing and legal contracts.
- **telephone: string:** Direct communication channel for notifications and booking management.

- **UniqueCode: string:** Internal key or member/customer number generated by the system to expedite searches in the database.

My opinion: Reflexive Relationship: A reflexive relationship (1...N to 1...N) is incorporated to model scenarios where a client can act as a guarantor, sponsor or dependent of other clients (for example, corporate or family accounts where a main

client manages or groups sub-clients).

Reservation Class

Purpose: Acts as the central transactional entity that links the customer, the agency, and the selected vehicles.

Attributes and justification:

- **StartDate: date and EndDate: date:** These define the rental period, which is essential for calculating vehicle availability.
- **RentalPrice: float:** Base rate applied during the rental period.
- **LitroGasolina: float:** Control of fuel level or consumption at the time of delivery or return, useful for additional charge policies.
- **TotalPrice: float:** Final calculated amount that includes rates, fuels and additional charges.

- **Delivered: boolean:** Boolean indicator (flag) that states whether the vehicle has already been physically delivered to the customer, facilitating operational control.

Car class (Vehicle)

Purpose: Models the inventory of cars available for rent in the fleet.

Attributes and justification:

- **LicensePlate: string:** License plate or license plate, used as a unique and unrepeatable physical identifier of the vehicle.
- **Model: string:** Specific car model.
- **Color: string:** Exterior color of the vehicle.
- **brand: string:** Car manufacturer's brand, essential attributes for the customer to identify and filter the vehicle of their preference.

Garage Class (Garage)

Purpose: Represents the physical infrastructure, storage center, workshop or parking lot where vehicles are kept when they are not in use or are receiving maintenance.

Attributes and justification:

- **Idgarage: string: Unique** identifier of the garage or shelter station.

Class Agency

Purpose: Represents the branch or administrative organization responsible for managing operations, contracts, and personnel.

Attributes and justification:

- **Idagency: string: Unique** identifier of the agency branch.

2. Justification of Relations and Multiplicities

- **Client to Reservation (1 to 1...N):** A client can make zero or multiple reservations over time (rental history), while each reservation must belong to a single registered client.
- **Agency a Reservation (1 to 0...N):** An agency manages operational reservations. A specific reservation is processed and managed by a specific agency, while an agency can manage multiple reservations or none at any given time.

- **Reservation a car (1...N to 1...N): Many-to-many** relationship justified because a reservation can include one or more vehicles (e.g., multiple rentals for fleets or groups), and a vehicle can be associated with different reservations at different times throughout its operational life cycle.
- **Car to Garage (0...N to 1):** A vehicle is assigned or stored in a specific garage at any given time, while a garage can house from zero to multiple vehicles depending on the capacity of its physical infrastructure.

3. Conclusion

Upon completion of the project, several questions arose when presenting it to the group, as various approaches were suggested. However, none were incorrect, since everyone interprets the result differently as long as the points outlined in the problem statement are followed, thus ensuring the model's coherence. During the process of creating the class diagram within the team, several ideas were proposed, with the issue of guarantors generating the most debate. Some members suggested creating a separate class for guarantors, arguing that the problem statement required one guarantor for each client. Under this proposal, this class would contain the same attributes as the client class, with the difference being that it would act as a guarantor, even though a guarantor remains a client of the company. This would establish a relationship where a client can have multiple guarantors and a guarantor can have multiple clients. This alternative generated confusion and doubts, as did several alternative scenarios, so the decision was ultimately made to resolve the issue through a reflexive relationship within the client class itself.

